

PW Skills: AWS Assignment Solutions

1. Explain the difference between AWS Regions, Availability Zones, and Edge Locations. Why is this important for data analysis and latency-sensitive applications?

Ans.

- AWS Regions: These are separate and independent large geographical areas where AWS deploys its cloud infrastructure. Each AWS region might contain at least three availability zones. Eg. us-east-1, ap-south-1
- Availability Zones (AZs): These are distinct, physically separate data centers located in a region equipped with their own power, cooling and networking system.
- Edge Locations: These are the individual sites used by Amazon Cloudfront, located around the world to store cache closer to the customers for faster delivery.

Importance:

- To comply with the organisational rules and data residency laws to store data locally or within a geographical location.
- To make data pipelines failure proof by storing your data in multiple AZs thus minimising downtime risks.
- Keeping data closer to customers reduces the latency as data stored far away takes higher time to load and comes with extra costs.
- Setting up edge locations as per the end users irrespective of storage regions helps to transfer and load data faster improving delivery and customer experience.

2. Using the AWS CLI, list all available AWS regions. Share the command used and the output.

Ans. Command used: `aws ec2 describe-regions --all-regions --output table`

```
C:\Users\mshub>aws ec2 describe-regions --all-regions --output table
```

DescribeRegions		
Regions		
Endpoint	OptInStatus	RegionName
ec2.ap-south-2.amazonaws.com	not-opted-in	ap-south-2
ec2.ap-south-1.amazonaws.com	opt-in-not-required	ap-south-1
ec2.eu-south-1.amazonaws.com	not-opted-in	eu-south-1
ec2.eu-south-2.amazonaws.com	not-opted-in	eu-south-2
ec2.me-central-1.amazonaws.com	not-opted-in	me-central-1
ec2.il-central-1.amazonaws.com	not-opted-in	il-central-1
ec2.ca-central-1.amazonaws.com	opt-in-not-required	ca-central-1
ec2.ap-east-2.amazonaws.com	not-opted-in	ap-east-2
ec2.mx-central-1.amazonaws.com	not-opted-in	mx-central-1
ec2.eu-central-1.amazonaws.com	opt-in-not-required	eu-central-1
ec2.eu-central-2.amazonaws.com	not-opted-in	eu-central-2
ec2.us-west-1.amazonaws.com	opt-in-not-required	us-west-1
ec2.us-west-2.amazonaws.com	opt-in-not-required	us-west-2
ec2.af-south-1.amazonaws.com	not-opted-in	af-south-1
ec2.eu-north-1.amazonaws.com	opt-in-not-required	eu-north-1
ec2.eu-west-3.amazonaws.com	opt-in-not-required	eu-west-3
ec2.eu-west-2.amazonaws.com	opt-in-not-required	eu-west-2
ec2.eu-west-1.amazonaws.com	opt-in-not-required	eu-west-1
ec2.ap-northeast-3.amazonaws.com	opt-in-not-required	ap-northeast-3
ec2.ap-northeast-2.amazonaws.com	opt-in-not-required	ap-northeast-2
ec2.me-south-1.amazonaws.com	not-opted-in	me-south-1
ec2.ap-northeast-1.amazonaws.com	opt-in-not-required	ap-northeast-1
ec2.sa-east-1.amazonaws.com	opt-in-not-required	sa-east-1
ec2.ap-east-1.amazonaws.com	not-opted-in	ap-east-1
ec2.ca-west-1.amazonaws.com	not-opted-in	ca-west-1
ec2.ap-southeast-1.amazonaws.com	opt-in-not-required	ap-southeast-1
ec2.ap-southeast-2.amazonaws.com	opt-in-not-required	ap-southeast-2
ec2.ap-southeast-3.amazonaws.com	not-opted-in	ap-southeast-3
ec2.ap-southeast-4.amazonaws.com	not-opted-in	ap-southeast-4
ec2.us-east-1.amazonaws.com	opt-in-not-required	us-east-1
ec2.ap-southeast-5.amazonaws.com	not-opted-in	ap-southeast-5
ec2.us-east-2.amazonaws.com	opt-in-not-required	us-east-2
ec2.ap-southeast-6.amazonaws.com	not-opted-in	ap-southeast-6
ec2.ap-southeast-7.amazonaws.com	not-opted-in	ap-southeast-7

3. Create a new IAM user with least privilege access to Amazon S3. Share your attached policies (JSON or screenshot).

Ans. A user created with only read access to the S3 bucket.

Users (1) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search

< 1 > ⚙

☐

User name

▲

Path

▼

Group: ▼

Last activity

▼

MFA

▼

Password age

▼

Console last sign-in

☐

[least_privilege_user1](#)

/

0

✔ 9 minutes ago

-

✔ 8 minutes

September 28, 2025

Permissions defined in this policy Info

Copy

Summary

JSON

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

1 {

2 "Version": "2012-10-17",

3 "Statement": [

4 {

5 "Effect": "Allow",

6 "Action": [

7 "s3:Get*",

8 "s3:List*",

9 "s3:Describe*",

10 "s3-object-lambda:Get*",

11 "s3-object-lambda:List*",

12],

13 "Resource": "*"

14 }

15]

16 }

4. Compare different Amazon S3 storage (Standard, Intelligent-Tiering, Glacier). When should each be used in data analytics workflows?

Ans.

- S3 Standard: It is used for high performance, frequently accessed and modified data. It has the lowest latency and is the costliest of the others. If you have raw data in your ML pipeline which is used to perform frequent EDAs and train or test the model it should be stored in S3 Standard.
- S3 Intelligent-Tiering: Automatically optimises data and moves them on the basis of usage patterns. If you don't know the usage pattern - how frequently the data will be used it is better to move it here.
- S3 Glacier: Data which is rarely used or accessed for active analysis but must be retained for future audit or compliance purposes. It is the cheapest category among all. Historical logs or old dataset can be stored here.

5. Create an S3 bucket and upload a sample dataset (CSV or JSON). Enable versioning and show at least two versions of one file.

temp-shubh-bucket1 Info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 > ⚙

☐

Name

▲

Type

▼

Last modified

▼

Size

▼

Storage class

▼

☐

[airline_monthly_data.csv](#)

csv

September 28, 2025, 21:28:47 (UTC+05:30)

25.6 KB

Standard

Buckets > temp-shubh-bucket1 > airline_monthly_data.csv

airline_monthly_data.csv Info

Copy S3 URI

Download

Open

Object actions

Properties

Permissions

Versions

Versions (2)

Download

Open

Delete

Actions

< 1 >

☐

Version ID

Type

Last modified

Size

Storage cl...

☐

tusdfE4JodQaoSC3FJ... (Current version)

csv

September 28, 2025, 21:28:47 (UTC+0...

25.6 KB

Standard

☐

0i9eMXyaYL8Odc1...

csv

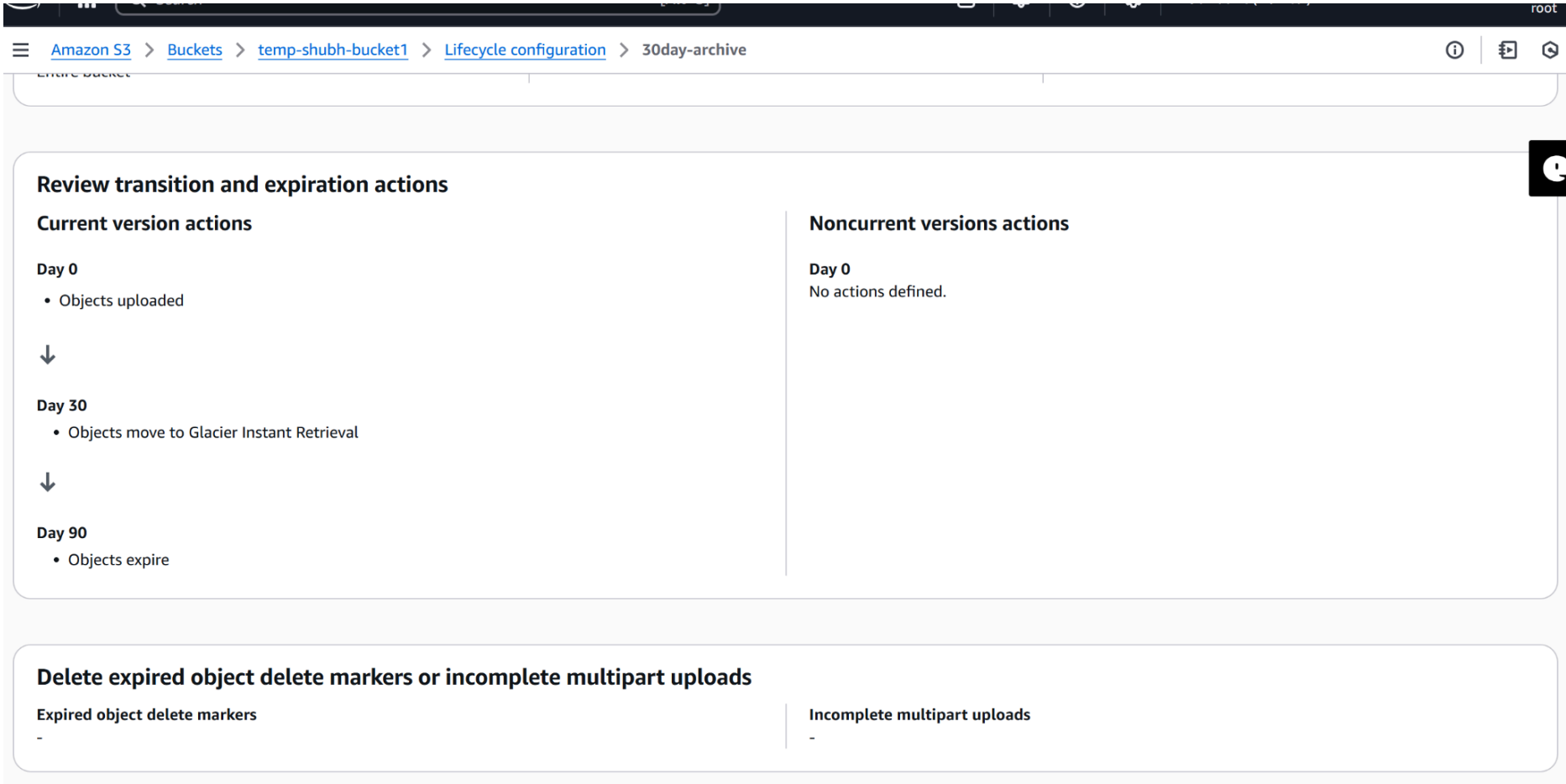
September 28, 2025, 21:20:16 (UTC+0...

37.7 KB

Standard

6. Write and apply a lifecycle policy to move files to Glacier after 30 days and delete them after 90. Share the policy JSON or Screenshot.

Ans.



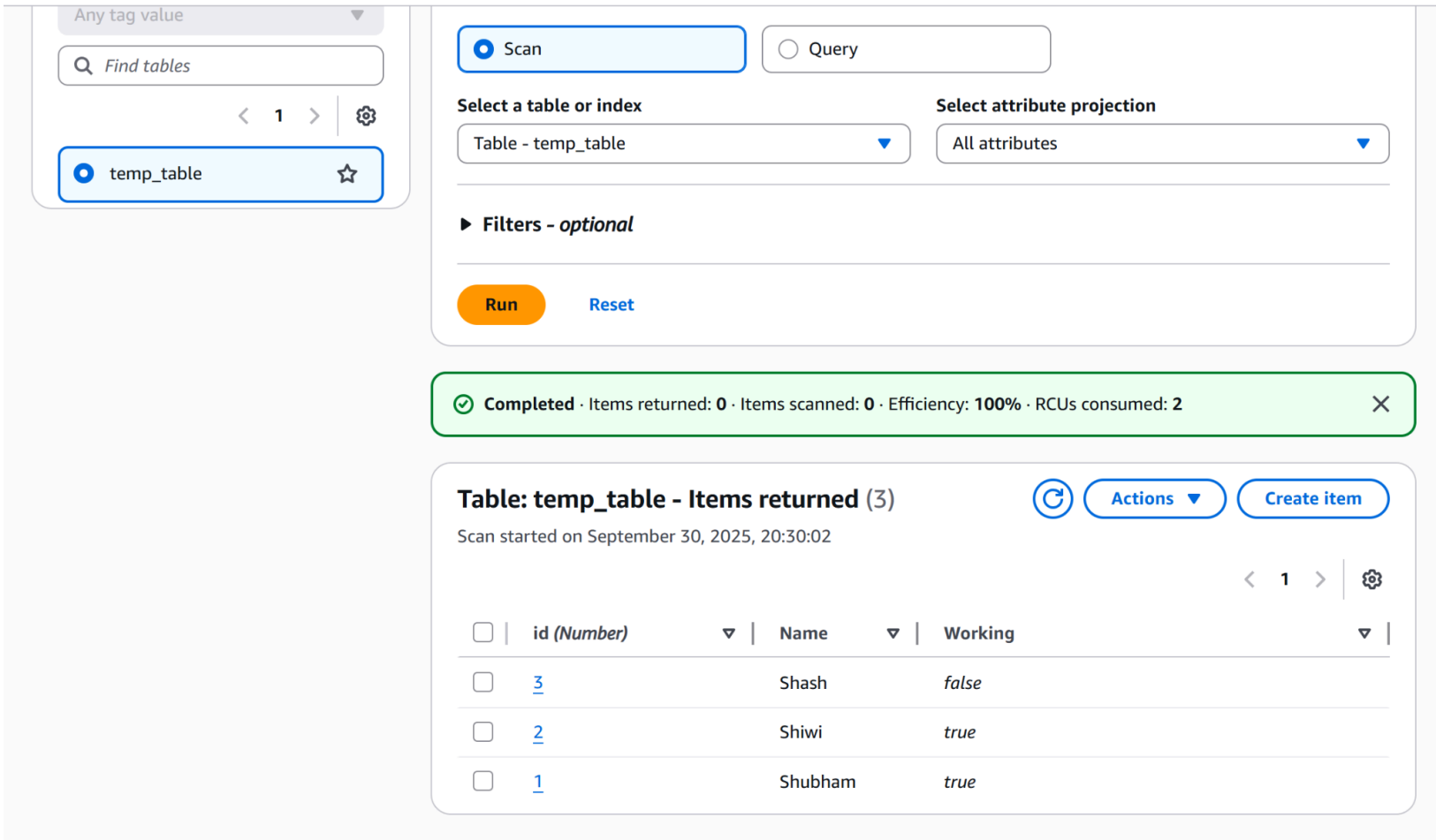
7. Compare RDS, DynamoDB, and Redshift for use in different stages of a data pipeline. Give one use case for each.

Ans.

- RDS: It is a service to set up, manage and scale relational databases in the AWS cloud. It takes care of database management and administration tasks such as data migration, recovery, backup and patching. Use case can be to manage databases that track pricing and inventories
- DynamoDB: It is a No-SQL database which supports key-value and document data structures. It is used for its performance, low latency and scalability. The use case is to store data for millions of shopping carts for a large number of people.
- Redshift: It is a data warehouse service provided by AWS to handle a large amount of data for analytical (OLAP) processing. Use case is to perform heavy analytical queries and dashboards after ETL process.

8. Create a DynamoDB table and insert 3 records manually. Then write a Lambda function that adds records when triggered by S3 uploads.

Ans.



```
1 import boto3
2 import json
3 import os
4
5 dynamodb = boto3.resource('dynamodb')
6 table = dynamodb.Table(os.environ['DDB_TABLE'])
7
8 def lambda_handler(event, context):
9     # Event contains S3 object info
10    for record in event['Records']:
11        s3 = record['s3']
12        bucket = s3['bucket']['name']
13        key = s3['object']['key']
14        size = s3['object']['size']
15
16        # create a simple record
17        item = {
18            'eventId': key,
19            'bucket': bucket,
20            'key': key,
21            'size': str(size),
22            'timestamp': record['eventTime']
23        }
24        table.put_item(Item=item)
25    return {'statusCode': 200}
26
```

9. What is serverless computing? Discuss pros and cons of using AWS Lambda for data pipelines.

Pros

Cons

10. Create a Lambda function triggered by S3 uploads that logs file name, size, and timestamp to Cloudwatch. Share code and a log screenshot.
Ans.

Lambda > Functions > s3_file_logger

s3_file_logger

ThrottleCopy ARNActions

✓ The trigger new-bucket-sm1 was successfully added to function s3_file_logger. The function is now receiving events from the trigger.

Function overviewInfo

Export to Infrastructure ComposerDownload

DiagramTemplate

s3_file_logger

Layers(0)

S3

+ Add trigger

+ Add destination

Description

-

Last modified

2 minutes ago

Function ARN

arn:aws:lambda:ap-south-1:792023347751:function:s3_file_logger

Function URL

Info

CodeTestMonitorConfigurationAliasesVersions

Lambda > Functions > s3_file_logger

EXPLORER

S3_FILE_LOGGER

lambda_function.py

DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [NONE SELECTED]

+ Create new test event

ENVIRONMENT VARIABLES

lambda_function.py

```
1 import logging
2
3 logger = logging.getLogger()
4 logger.setLevel(logging.INFO)
5
6 def lambda_handler(event, context):
7     for record in event['Records']:
8         s3 = record['s3']
9         bucket = s3['bucket']['name']
10        key = s3['object']['key']
11        size = s3['object'].get('size', 'unknown')
12        ts = record.get('eventTime')
13        logger.info(f"S3 object uploaded: bucket={bucket}, key={key}, size={size}, ts={ts}")
14    return {"status": "ok"}
15
16
```

Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)

Log events

ActionsStart tailingCreate metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

1m1hUTC timezoneDisplay

Timestamp

Message

No older events at this moment. [Retry](#)

2025-09-30T15:51:30.124Z

INIT_START Runtime Version: python:3.13.v60 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:6b9e6bdace9349e61d8...

2025-09-30T15:51:30.212Z

START RequestId: 204fe90b-bbb3-430e-92e3-0e3077d88517 Version: \$LATEST

2025-09-30T15:51:30.213Z

[INFO] 2025-09-30T15:51:30.213Z 204fe90b-bbb3-430e-92e3-0e3077d88517 S3 object uploaded: bucket=new-bucket-sm1, key=fil...

2025-09-30T15:51:30.221Z

END RequestId: 204fe90b-bbb3-430e-92e3-0e3077d88517

2025-09-30T15:51:30.221Z

REPORT RequestId: 204fe90b-bbb3-430e-92e3-0e3077d88517 Duration: 2.18 ms Billed Duration: 87 ms Memory Size: 128 MB Max...

No newer events at this moment. Auto retry paused. [Resume](#)

Ans.

csv_2_parquet

Last modified on 10/2/2025, 3:29:17 PM

ActionsSaveRun

ScriptJob detailsRunsData qualitySchedulesVersion Control

ScriptInfo

1import sys

2from awsglue.context import GlueContext

3from pyspark.context import SparkContext

4from awsglue.transforms import *

5from awsglue.utils import getResolvedOptions

6

7args = getResolvedOptions(sys.argv, ['JOB_NAME'])

8sc = SparkContext()

9glueContext = GlueContext(sc)

10spark = glueContext.spark_session

11

12# read CSV from S3 directly

13df = spark.read.option("header", "true").csv("s3://temp-shubh-bucket1/raw/")

14

15

16# write as partitioned parquet

17df.write.mode("overwrite").parquet("s3://temp-shubh-bucket1/output/")

18

PythonLn 1, Col 1Errors: 0Warnings: 0

> Buckets> temp-shubh-bucket1> output/

output/

Copy S3 URI

Objects

Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 >

☐	Name	Type	Last modified	Size	Storage class
☐	part-00000-c44aeb27-d00c-4b62-85a0-14f41f387fa5-c000.snappy.parquet	parquet	October 2, 2025, 15:30:29 (UTC+05:30)	728.0 B	Standard

12. Explain the difference between Kinesis Data Streams, Kinesis Firehose, and Kinesis Data Analytics. Provide a real-world example of how each would be used.

Ans.

- Kinesis Data Streams: Real time, highly durable stream of raw data that will be processed further. It stores all the data coming and can clean and process it independently and concurrently. Usage: Real time clickstream information for e-commerce websites utilizes Data Steams to ingest the data.
- Kinesis Firehouse: This is a fully automated service used to load and transfer data for long term storage into S3, Redshift, ElasticSearch etc. Usage: Loading data into data lakes (S3) or data warehouses (Redshift) for search/analytics without writing any code.
- Kinesis Data Analytics: To perform real time analytics on the continuously streaming data using Standard SQL or Apache Flink. Usage: Real time insights and aggregation on the streaming data and publish alerts for anomalies.

13. What is columnar storage and how does it benefit Redshift performance for analytics workloads?

Ans. Columnar storage is a database management system approach used by services like AWS Redshift that stores data column by column unlike traditional database management systems which store row-by-row. The benefit of this approach is that it makes querying incredibly fast for analytics as it only reads the data that is needed, compresses the data efficiently and processes the data in large, optimised batches.

14. Load a CSV file from S3 into Redshift using the COPY command. Share table schema, command used, and sample output from a query.

Ans. Cannot perform the task completely as it needs a subscription

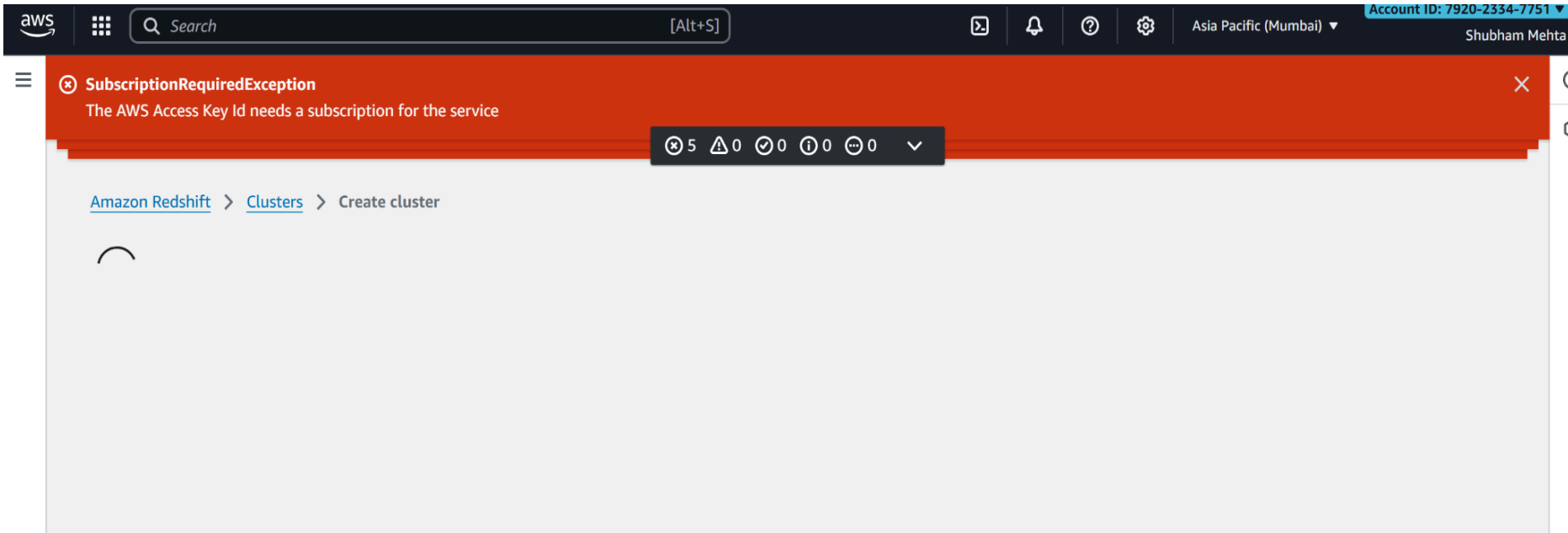


Table Schema:

Column Name	Data Type	Constraints	Encoding	Notes
transaction_id	INT	NOT NULL	ZSTD	Also used as DISTKEY
item_name	VARCHAR(255)		ZSTD	Stores the product/item name
quantity	INT		ZSTD	Number of units sold
sale_date	DATE		ZSTD	Also used as SORTKEY

```
-- COPY command to load data from S3 into the Redshift table
COPY public.sales_transactions
FROM s3://[YOUR-S3-BUCKET-NAME]/data/sales/sales_data.csv
IAM_ROLE [YOUR-IAM-ROLE-ARN]
FORMAT AS CSV
DELIMITER ','
IGNOREHEADER 1
DATEFORMAT 'YYYY-MM-DD'
MAXERROR 5;
```

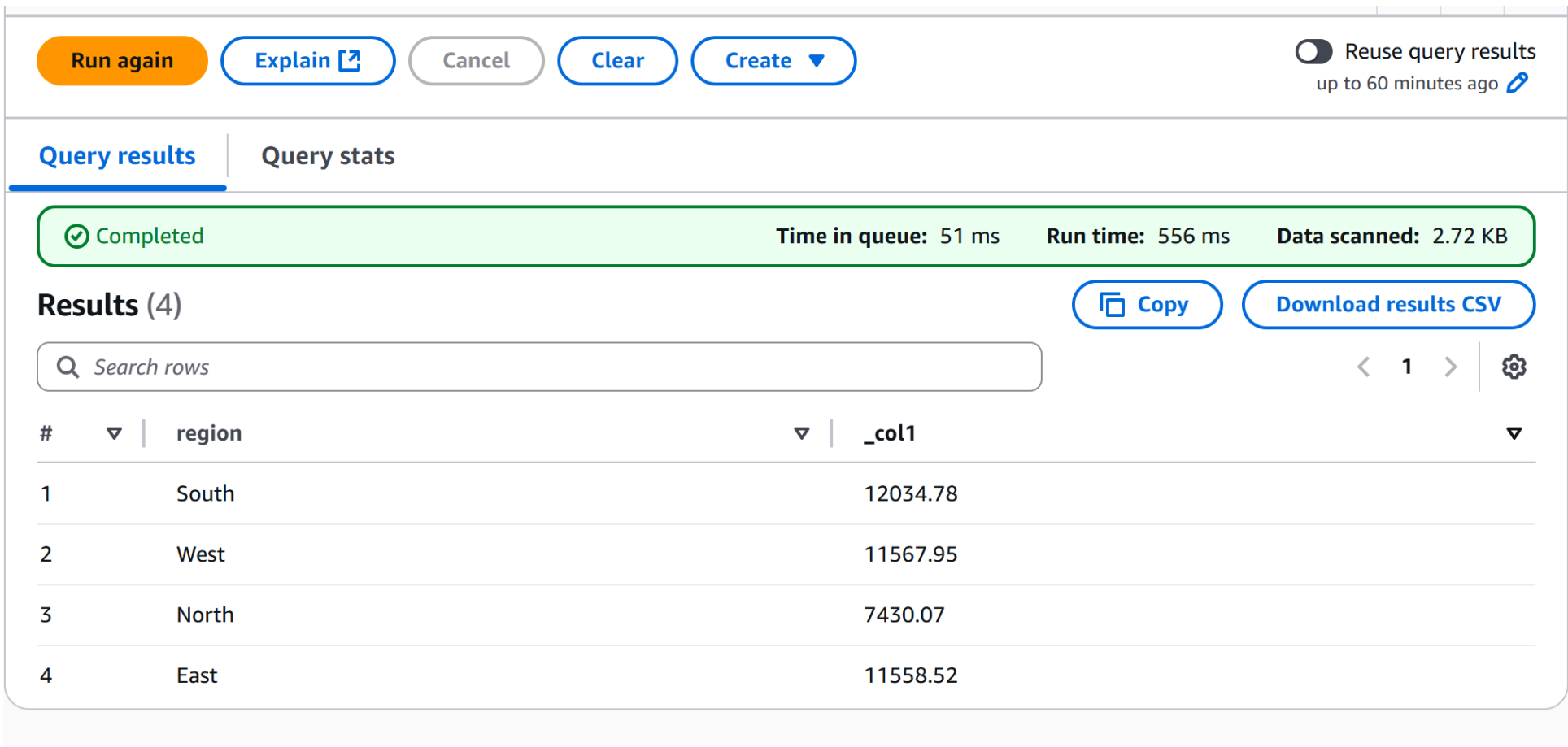
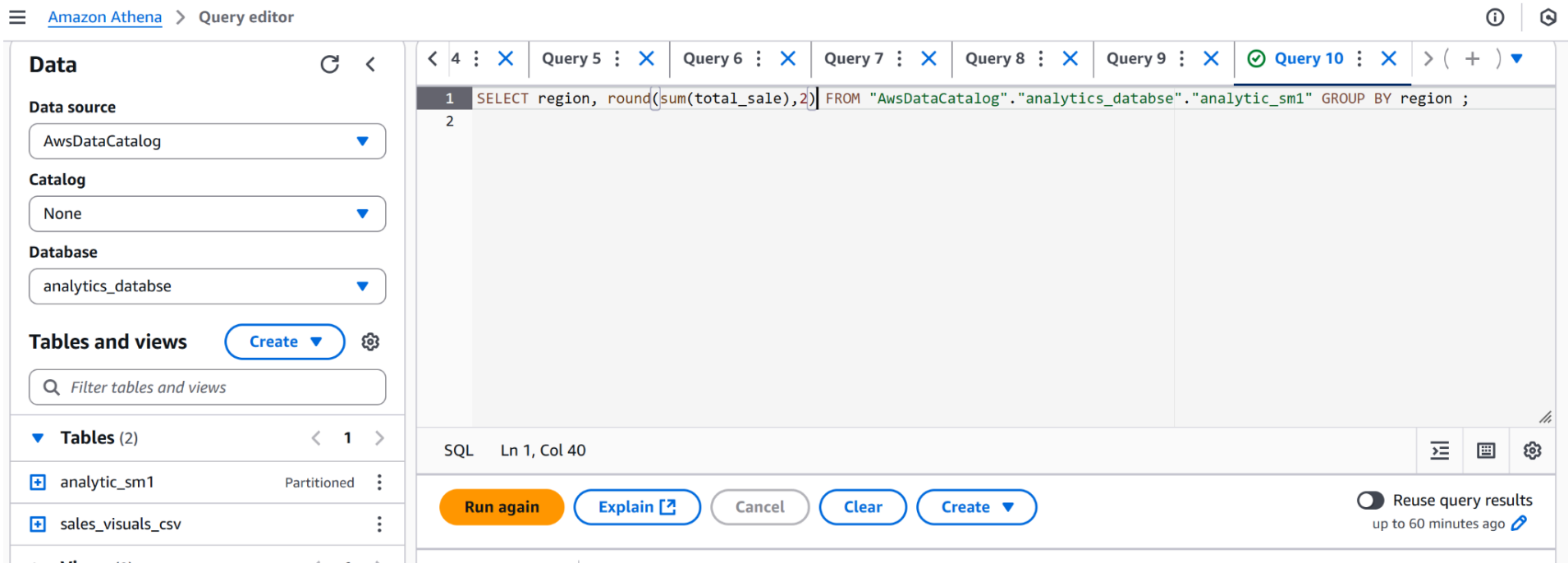
15. What is the role of the AWS Glue Data Catalog in Athena? How does schema-on-read work?

Ans. The AWS Glue Catalog serves as a central and unified metadata repository for your entire AWS data estates. All your data files stored in S3 are cataloged by AWS Glue and store all the information regarding the data: schema, location, format, partitioning etc. So when Athena is used to call the data or query, the AWS Glue Data Catalog acts as an instruction manual that tells Athena how to query the data stored in the data lake.

Traditional databases use schema on write approach while storing any data where data is first transformed into predefined schema structure and then stored. While Athena uses schema-on-read approach where the data is transformed while reading or querying using metadata from Glue Data Catalog making query faster and flexible as per need.

16. Create an Athena table from S3 data using Glue Catalog. Run a query and share the SQL + result screenshot.

Ans.



17. Describe how Amazon Quicksight supports business intelligence in a serverless data architecture. What are SPICE and embedded dashboards?

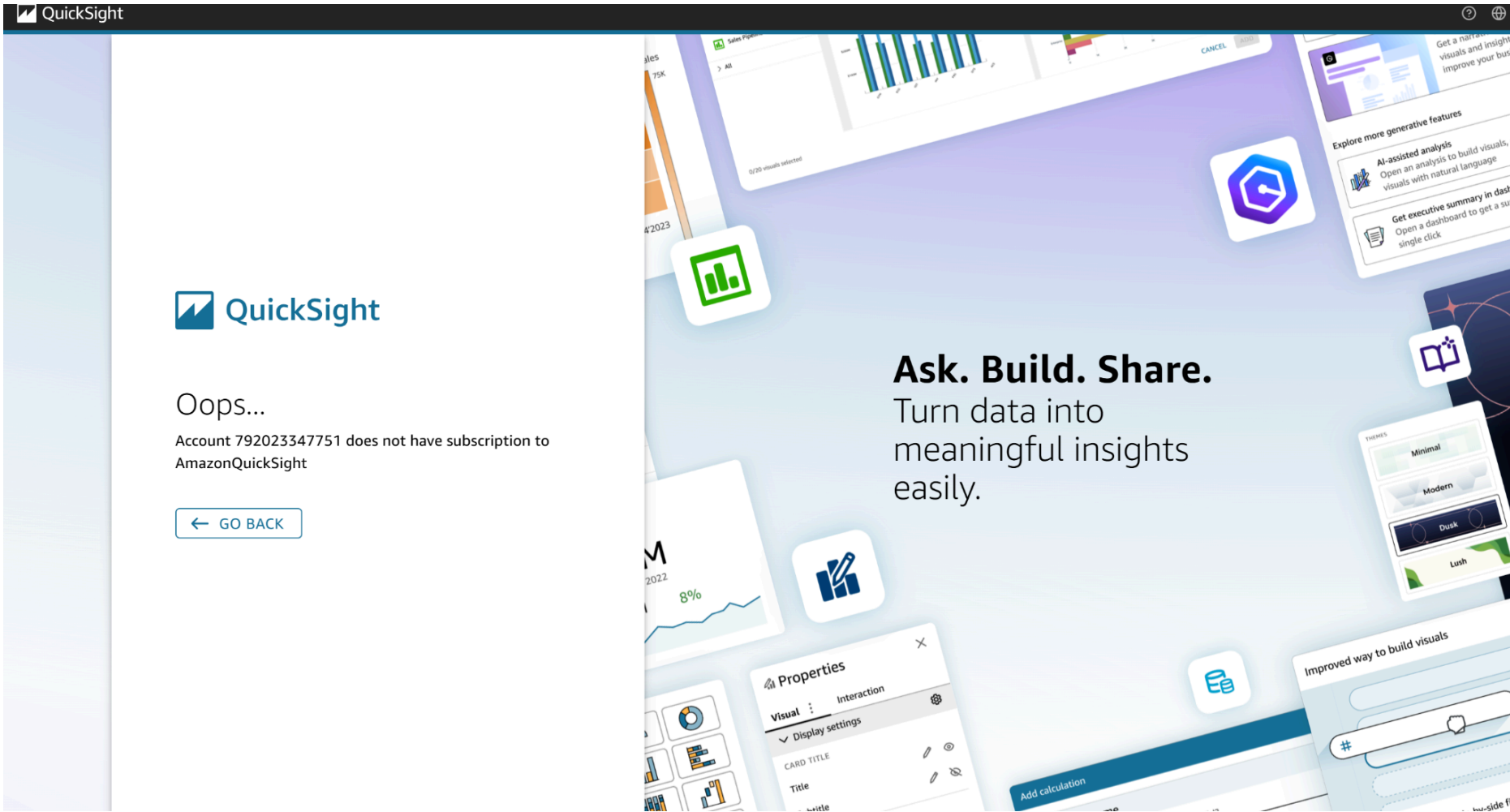
Ans. Quicksight is a business intelligence serverless platform by AWS that helps in building visualizations or dashboards for the end-users. It can natively connect with services such as Athena, Redshift or Kinesis to directly make dashboards and scales automatically for users and dashboards.

SPICE: Super-fast Parallel In-memory Calculation Engine is Quicksight's in memory storage and engine which loads and stores cached data for faster visualizations.

Embedded Dashboards: The embedded dashboards integrates Quicksight's visualizations on a website or web application directly to view and interact with dashboards without actually opening Quicksight's each time separately

18. Connect Quicksight to Athena or Redshift and build a dashboard with at least one calculated field and one filter. Share a screenshot of your final dashboard.

Ans. Quicksight is not available in free-tier anymore after July 2025.



19. Explain how AWS CloudWatch and CloudTrail differ. In a data analytics pipeline, what role does each play in monitoring, auditing, and troubleshooting?

Ans. Both CloudWatch and CloudTrail deal with logs and monitoring but look after different aspects of an AWS service. CloudWatch is used to observe and monitor the different functions and operations such as CPU usage, job status and errors, lambda errors, and dashboards. CloudTrail is used for auditing and compliance purposes like how did changes to what and tracking all the changes to the resources.

In a data analytics pipeline AWS CloudWatch monitors all the runtime metrics: Glue job process duration, storing logs, lambda errors, queue time. While AWS CloudTrail records and audits all the administrative processes: IAM Policy changes, tracking changes in resources, logging API calls.

20. Describe a complete end-to-end data analytics pipeline using AWS services. Include services for data ingestion, storage, transformation, querying, and visualization. (Example: S3 → Lambda → Glue → Quicksight). Explain why you would choose each service for the stage it's used in.

Ans.

Stage	AWS Service	Role in the Pipeline	Why This Service?
1. Ingestion (Real-time/Streaming)	Kinesis Data Firehose	Captures real-time streaming data (e.g., website clicks, IoT telemetry) and automatically loads it into Amazon S3.	It is a fully managed, serverless service that handles buffering, compression, and encryption automatically. It scales effortlessly to handle spikes in traffic without manual intervention.
1. Ingestion (Batch/Event-Driven)	AWS Lambda	Triggers upon the availability of new batch files (e.g., daily CSV uploads) in a landing S3 bucket to move or process the data.	It is a serverless compute service used for lightweight, event-driven tasks. It runs code (like Python scripts) to validate incoming files or kick off the next stage, paying only for execution time.

2. Storage (Data Lake Foundation)	Amazon S3 (Simple Storage Service)	Stores the raw ingested data (Landing Zone) and the processed, structured data (Curated Zone) in open formats like Parquet or ORC.	It offers industry-leading durability (99.999999999%) and virtually unlimited scalability at a very low cost. It is the fundamental building block for a data lake due to its native integration with all other AWS analytics services.
3. Cataloging & Metadata	AWS Glue Data Catalog	Serves as the central metadata repository for the data lake, storing table definitions, schemas, and partition information for data stored in S3.	It is a managed, persistent metadata store that allows querying services (like Athena and Redshift Spectrum) to understand the structure of the data in S3 without needing to move it.
4. Transformation (ETL/ELT)	AWS Glue (ETL Jobs)	Runs serverless Spark jobs to perform complex Extract, Transform, and Load (ETL) operations, such as cleaning, normalizing, joining datasets, and converting data to optimized columnar formats (Parquet).	It is a fully managed, serverless data integration service. It automatically provisions, configures, and scales the compute resources needed for Spark processing, simplifying ETL pipeline management.
5. Querying & Analysis (Ad-hoc)	Amazon Athena	Allows analysts and data scientists to run standard SQL queries directly against the data stored in S3 (using the Glue Data Catalog metadata).	It is a serverless, pay-per-query service. It is ideal for interactive, ad-hoc analysis because there is no infrastructure to manage, and queries are typically very fast due to reading optimized formats like Parquet.
6. Visualization & Reporting	Amazon QuickSight	A business intelligence (BI) service used to create and publish interactive dashboards, reports, and visualizations from data analyzed by Athena.	It is a scalable, serverless, and machine-learning-powered BI tool. Its integration with native AWS data sources (like Athena and S3) is seamless, and it includes SPICE (its in-memory engine) for fast, concurrent access to complex datasets.